

Nuclear Stability

What is the nuclear stability? Nuclear stability means that nucleus is stable meaning that it does not spontaneously emit any kind of radioactivity (radiation). On the other hand, if the nucleus is unstable (not stable), it has the tendency of emitting some kind of radiation, i.e., it is radioactive. Therefore the radioactivity is associated with unstable nucleus:

Stable nucleus – non-radioactive

Unstable nucleus – radioactive

Keep in mind that less stable means more radioactive and more stable means less radioactive.

We want to know why there is a radioactivity. What makes the nucleus a stable one? There are no concrete theories to explain this, but there are only general observations based on the available stable isotopes. It appears that neutron to proton (n/p) ratio is the dominant factor in nuclear stability. This ratio is close to 1 for atoms of elements with low atomic number and increases as the atomic number increases. Then how do we predict the nuclear stability? One of the simplest ways of predicting the nuclear stability is based on whether nucleus contains odd/even number of protons and neutrons:

Protons	Neutrons	Number of Stable Nuclides	Stability
Odd	Odd	4	least stable ↓ most stable
Odd	Even	50	
Even	Odd	57	
Even	Even	168	

- Nuclides containing odd numbers of both protons and neutrons are the least stable means more radioactive.
- Nuclides containing even numbers of both protons and neutrons are most stable means less radioactive.
- Nuclides contain odd numbers of protons and even numbers of neutrons are less stable than nuclides containing even numbers of protons and odd numbers of neutrons.

In general, nuclear stability is greater for nuclides containing even numbers of protons and neutrons or both.

Example

Based on the even-odd rule presented above, predict which one would you expect to be radioactive in each pair?

